17 The General Energy Balance Equation

Why are we learning this?

So far in this course, we have primarily focused on mole balances and reaction rates. However, chemical reactions do more than consume reactants and produce products—they also release or absorb energy. Because reaction rates depend strongly on temperature, energy changes can dramatically affect reactor performance.

By the end of today's lesson, you should understand:

- Why energy balances are important in reactor design.
- How chemical reactions store and release energy.
- The terms that appear in the General Energy Balance.
- Why heats of formation must be included in reactor energy balances.
- How energy balances complement mole balances.
- Why temperature becomes a critical reactor design variable.

Course Roadmap

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General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)

↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
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↓
Catalytic Reactions
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One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
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Reactor Safety
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Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Mole Balances Tell Us: How much reaction occurs.

Energy Balances Tell Us: How temperature changes.

Both are required for realistic reactor design.

Energy balances are **key** to chemical reactor design. Chemical bonds store **a lot** of energy; that's why breaking and forming bonds in chemical reactions results in large ΔH^{rxn}.

To get a feel for what I mean by large, consider the amount of energy needed to vaporize a mole of water … ~ 40 kJ/mol water. Now consider heat released due to the reaction H_2 (g) + 1/2 O_2 (g) --> H_2O (g) … ~ 240 kJ/mol water.

The reason for the difference is the relative strengths of the bonds being broken and formed: Physical bonds between water molecules in the first case versus chemical bonds between the atoms in a molecule in the second. **Chemical bonds store <u>a lot</u> of potential energy.**

Why Are Reaction Energies So Large?

Chemical bonds store large amounts of potential energy. When reactions occur:

- Existing bonds are broken.
- New bonds are formed.

The difference in bond energies appears as heat absorbed or released. This is why chemical reactions often release far more energy than ordinary physical processes such as vaporization.

Of course, not all reactions give off as much heat as the H_2 (g) + 1/2 O_2 (g) --> H_2O (g) reaction … The heats of some reactions, e.g., isomerization reactions, are a lot lower, e.g., 40 kJ/mol. However, given the number of chemical species in a reactor, we **need to concern ourselves with all reactions that <u>could</u> happen**, not just those that are desired/designed for.

As we move through this unit on energy balances, we will start by learning how to derive energy balances for common reactors, then we will use the energy balances to study non-isothermal reactor design, then we will examine "reaction runaway," which is often caused when reactions other than those that are desired/designed for occur in a chemical reactor.

Why Does This Matter for Reactor Design?

A reactor may contain numerous possible reactions.

Some desired.

Some undesired.

Some highly exothermic.

Some mildly exothermic.

Even small amounts of unexpected chemistry can produce significant temperature changes.

Because reaction rates depend on temperature, energy effects can strongly influence reactor performance.

To start, consider a system (e.g., a CSTR, or a volume element in a PFR, i.e., something where the contents are well-mixed):

General form of the energy balance:

Rate of accumulation of energy within the system = rate of flow of heat into the system from the surroundings + rate of work done on the system by the surroundings + rate of energy added to the system by mass flow into the system - rate of energy leaving the system by mass flow out of the system.

Symbolically: $\frac{d\hat{E}}{dt} = \dot{Q} + \dot{W}_s + \sum_i F_{i0}H_{i0} - \sum_i F_i H_i$.

Energy Balances Are Also Accounting Equations

Just like mole balances, energy balances are conservation statements.

The question is no longer: Where did the molecules go?

Instead, we ask: Where did the energy go?

Every source and destination of energy must be accounted for.

Reading the Energy Balance

The General Energy Balance has four major contributions:

Heat Transfer + Work + Energy In With Flow - Energy Out With Flow

The difference between these terms determines whether energy accumulates within the system.

This is directly analogous to how mole balances track material entering, leaving, and reacting.

The *H*'s are **heats of formation**. E.g., they can be looked up on the NIST website: https://webbook.nist.gov/chemistry/.

Technically, the *H*'s are heats of formation at the reaction temperature. HOWEVER, while the heat of formation for an individual species varies significantly as a function of temperature, ΔH^{rxn} does not vary much at all with reaction temperature. Hence**, it is more important that you obtain a heat of formation for the energy balance than it is to adjust the heat of formation to the**

reaction temperature. In fact, a key mistake that students make is to only include the sensible heat changes for each component in their energy balances, instead of including the heat of formation, which includes all of the energy stored within the chemical bonds. Do not make this mistake! Always use the heat of formation, and always check the value that you use against the standard value (which is tabulated).

Key point: You must use the heat of formation in your energy balance; otherwise, you do not account for the energy that is released from the chemical bonds as they break and form.

For a visual on how the energy balance is derived, see the Interactive Computer Games (http://www.umich.edu/~elements/5e/icm/index.html) called "Heat Effects 1" and "Heat Effects 2".

Why Use Heats of Formation?

The heat of formation includes the chemical energy stored in molecular bonds. Without heats of formation:

Bond Breaking Energy + Bond Formation Energy

would not be accounted for.

As a result, the energy balance would miss one of the largest sources of energy change in a chemical reactor.

Common Energy-Balance Mistake

Many students include only sensible heat effects and forget heats of formation. This omission ignores the energy stored in chemical bonds. Always ask yourself:

Did I include heats of formation?

If not, the energy balance is likely incomplete.

At steady state, the energy balance becomes:

Implications on reactor design: The following quantities depend on temperature:

These equations need to be accounted for in non-isothermal reactor design problem solutions.

Why Temperature Changes Everything

Temperature influences:

- Rate constants
- Equilibrium constants
- Concentrations
- Physical properties

As a result:

Temperature
↓
Changes Reaction Behavior
↓
Changes Reactor Performance

This coupling is the central challenge of non-isothermal reactor design.

Summary and Key Takeaways

Today we introduced the General Energy Balance and its role in reactor design.

Key Ideas

✓ Chemical reactions can release or absorb large amounts of energy because chemical bonds store significant energy.

✓ Energy balances are conservation equations that account for all energy entering, leaving, and accumulating within a system.

✓ Heat transfer, work, and energy carried by flowing streams all appear in the energy balance.

✓ Heats of formation must be included to account for energy changes associated with chemical reactions.

☑ Ignoring heats of formation is one of the most common reactor energy-balance mistakes.

☑ Temperature strongly influences reaction rate and reactor performance.

Most Important Idea

Chemical Reactions
↓
Change Temperature
↓
Temperature Changes Rate
↓
Rate Changes Reactor Performance

This feedback loop is why energy balances are a fundamental part of chemical reactor design.